

Machine learning-guided analysis of CIGS solar cell efficiency: Deep learning classification and feature importance evaluation

A. Maoucha^a, T. Berghout^b, F. Djeflal^{a,*}, H. Ferhati^{a,c}

^a LEA, University of Batna2, Batna 05000, Algeria

^b University of Batna 2, Batna 05000, Algeria

^c ISTA, University of Larbi Ben M'hidi, Oum El Bouaghi, Algeria

ARTICLE INFO

Keywords:

CIGS
Solar cell
Machine Learning
Deep Learning
Classification

ABSTRACT

The increasing sensitivity of thin-film solar cells to variations in design parameters is becoming more pronounced with ongoing advancements in material science and device engineering. Even minor deviations in these parameters can significantly impact the performance of the solar cell devices. Therefore, an in-depth investigation of such variations is crucial to advancing the efficiency and reliability of thin-film solar cell technology. This paper presents an innovative design methodology by combining numerical simulations with Machine Learning (ML) techniques to explore and analyze the critical design parameters of CIGS thin-film solar cells. Specifically, accurate numerical simulations, incorporating defects and appropriate charge transport mechanisms in different layers, are utilized to simulate the current–voltage (I–V) characteristics. The study thoroughly examines the effects of design parameter variations and the role of high-efficiency absorber and buffer layer materials on energy conversion performance. Several ML algorithms are employed to evaluate and classify the most influential design parameters impacting key figures of merit, such as power conversion efficiency (PCE), open-circuit voltage (V_{OC}), and short-circuit current density (J_{SC}). The results highlight the significant impact of the buffer and absorber layers properties on the overall efficiency. These parameters are also crucial for predicting and optimizing PCE. Furthermore, ML models are leveraged to identify the optimal design parameters for maximizing each Figure-of-merit (FoM).

1. Introduction

In the 21st century, the transition toward renewable energy sources, as substitutes for fossil fuels, is recognized as one of the most pressing scientific and technological challenges [1–3]. The main objective of utilizing renewable energy is to mitigate air pollution caused by fossil fuel consumption, reduce global warming, and preserve the Earth's biochemical cycles. A sustainable energy source that maintains ecological balance is essential, and these are collectively known as renewable energy sources. Solar energy stands out as one of the most viable renewable resources due to its non-toxic, cost-effective, and environmentally friendly nature, positioning it as a promising alternative to meet contemporary energy demands [1–3]. To address the crystalline silicon (Si) solar cells limitations [4], thin-film (TF) technologies employing direct bandgap semiconductors, such as Cu(In,Ga)(S,Se)_2 (CIGS), GaAs, and CdTe, have been developed as cost-effective alternatives to Si-based cells. However, the presence of toxic elements like Cd

and As in some of these semiconductors poses challenges for their widespread commercial use [5–7]. As a result, research is increasingly focused on high-performance and appropriate materials for thin-film solar cell fabrication. Among these, thin CIGS (Cu(In,Ga)(S,Se)_2) solar cells have emerged as one of the most promising candidates. CIGS boasts a direct bandgap tunable between 1.0 and 1.7 eV, which is close to the ideal range for solar cell applications, alongside high absorption coefficient and p-type conductivity [5]. The versatility of CIGS in different stoichiometric compositions allows for effective bandgap tuning, enabling the optimization of solar cell performance without the need for toxic elements. Nevertheless, challenges such as lattice mismatching, recombination losses, and band-alignment issues still limit their performance [5–7]. Consequently, further performance improvement is required, where the development of new design and elaboration methodologies is becoming a new research trend to overcome these challenges. In this regard, selecting the correct buffer and optical window layers is of utmost importance when dealing with CIGS as an absorber

* Corresponding author.

E-mail address: faycal.djeflal@univ-batna2.dz (F. Djeflal).

<https://doi.org/10.1016/j.solener.2025.113251>

Received 12 September 2024; Received in revised form 31 December 2024; Accepted 2 January 2025

Available online 6 January 2025

0038-092X/© 2025 International Solar Energy Society. Published by Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

layer. A selection can be made based on two factors: one is band-offset alignment and the other one is the mismatch in lattice between different layers [8]. Additionally, the use of Cd in the traditional CdS buffer layer and indium in the ITO window layer presents environmental and economic limitations [8–11]. Consequently, there is a growing focus on optimizing CIGS solar cell structures by selecting alternative, non-toxic buffer layers (such as ZnS, ZnSe, InS, or SiC) and exploring cost-effective window layers to enhance device performance [8–10]. To overcome these limitations and further improve device performance and efficiency, researchers developed new numerical models and simulation approaches to optimize solar cell parameters and assess the impact of various material choices. Technology Computer-Aided Design (TCAD) tools are widely used to model CIGS thin-film solar cells (TFSCs) and explore ways to improve efficiency [12,13]. Additionally, machine learning (ML) techniques have emerged as valuable tools for optimizing design parameters, accelerating the discovery of new materials, and predicting device performance with reduced computational costs [14–16]. By integrating TCAD simulations with ML-based modeling, it becomes possible to evaluate the effects of critical design parameters and identify the optimal configurations for improved performance. In our design methodology, ML can provide valuable predictions regarding the performance, behavior, and characteristics of CIGS TFSCs. By training ML models, such as neural networks and support vector machines, on simulation data, we can predict electrical characteristics like open circuit voltage, short circuit current, Fill Factor and solar cell power efficiency [17]. These models offer rapid insights into how design changes impact device performance, minimizing the need for extensive physical simulations. Additionally, they help identify which design parameters most significantly affect solar cell performance, allowing us to focus efforts on the critical aspects of the design process. The integration of TCAD tools with ML provides a synergistic approach to overcoming the challenges of designing high-performance TFSCs. TCAD offers detailed physics-based simulations, while ML enhances predictive capabilities, optimizes design processes, classifies parameters, and accelerates simulations.

To identify potential design issues and gain a comprehensive understanding of the key parameters that influence the performance of CIGS TFSCs, this work combines numerical modeling using SCAPS simulations and machine learning (ML) techniques. The main objective is to develop a novel design framework that integrates numerical simulations with ML modeling to optimize CIGS TFSC structure using the appropriate key design parameters. This approach involves simulating the device to evaluate the impact of various design parameters on critical characteristics such as power conversion efficiency (PCE), open-circuit voltage (V_{OC}), and short-circuit current density (J_{SC}). These predictions are essential for assessing how the solar cell will perform under different operating conditions. By leveraging ML techniques, we analyze the prepared numerical dataset to train models that can predict device performance based on variations in key design parameters, including CIGS layer thickness, CIGS acceptor density, buffer layer thickness, buffer layer donor density, buffer layer material and optical window thickness. The prediction and classification results will help identify the most influential design parameters and define the optimal solar cell structure. Moreover, this approach minimizes the need for extensive device numerical simulations of thin-film multilayer structures by using ML-based modeling to predict the effects of parameter variations, significantly streamlining the design process.

2. Solar cell structure and modeling methodology

The solar cell design presented incorporates a CIGS absorber layer deposited onto soda lime glass substrates with a Mo coating at the base of the cell. A buffer layer is applied atop the absorber layer, as depicted in Fig. 1. The uppermost portion features front contact fingers formed from aluminum-doped ZnO (AZO), a transparent conductive oxide (TCO) known for its low cost, non-toxicity, and enhanced durability

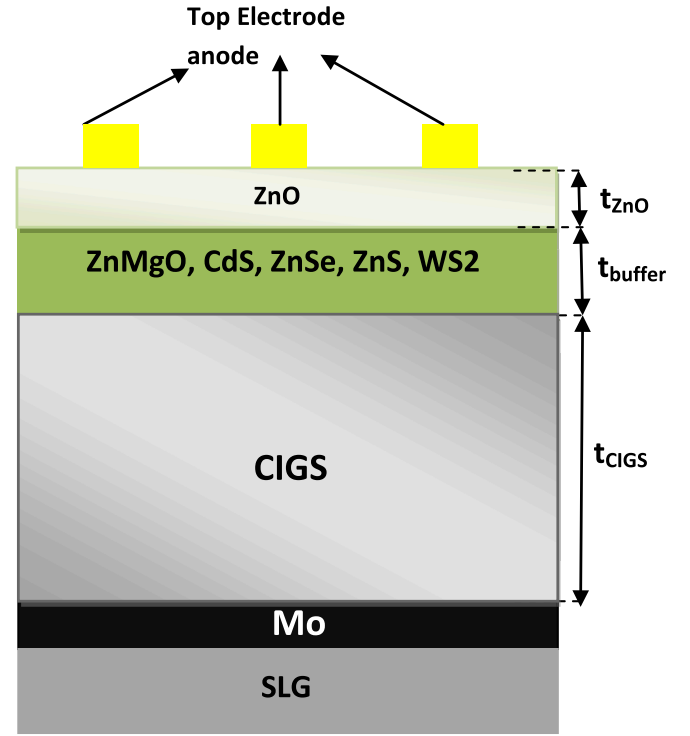


Fig. 1. Cross-sectional view of a CIGS solar cell.

compared to the widely used tin-doped indium oxide (ITO) [18–20]. In this Figure, t_{ZnO} , t_{buffer} and t_{CIGS} are the ZnO, buffer layer and CIGS thicknesses respectively.

The SCAPS-1D software was used to simulate the device configurations, this software was developed by Prof. Marc Burgelman and his team at the University of Gent [21,22]. Various generations of solar cell structures can be simulated by this tool, which offers versatility in simulating and accurately predicting of their performance under different conditions such as defect density, absorption characteristics, carrier concentration and bandgap grading. Performance parameters, including photogenerated current density, open circuit voltage, fill factor, and power conversion efficiency, are calculated by solving Poisson's equation and the carrier continuity equations through numerical differentiation [22]. SCAPS computations are based on core principles, specifically Poisson's equation, the carrier continuity equation, and the drift-diffusion equation [23].

$$\frac{dp_n}{dt} = G_p - \frac{p_n - p_{n0}}{\tau_p} - p_n u_p \frac{dE}{dx} - u_p E \frac{dp_n}{dx} + D_p \frac{d^2 p_n}{dx^2} \quad (1)$$

$$\frac{dn_p}{dt} = G_n - \frac{n_p - n_{p0}}{\tau_n} - n_p u_n \frac{dE}{dx} - u_n E \frac{dn_p}{dx} + D_n \frac{d^2 n_p}{dx^2} \quad (2)$$

$$\frac{d}{dx} \left(\epsilon(x) \frac{d\phi}{dx} \right) = q[p(x) - n(x) + N_{d^+}(x) - N_{a^-}(x) + p_t(x) - n_t(x)] \quad (3)$$

The simulations are performed at a stable temperature of 300 K. Illumination is applied through the front contact, using the Air Mass 1.5 global spectrum at a light intensity equivalent to one sun (1000 W/m²). To assess the accuracy of the simulation, we have compared the current-voltage characteristics generated by the SCAPS-1D against experimental data [24], as illustrated in Fig. 2(a). The comparison reveals a strong correlation between the simulated and experimental results, confirming the simulation's reliability. Additionally, the findings indicate that SCAPS-1D is capable of accurately modeling the electrical and optical behavior of solar cells, where the simulated I-V characteristics closely match the experimental outcomes, as evidenced by the excellent

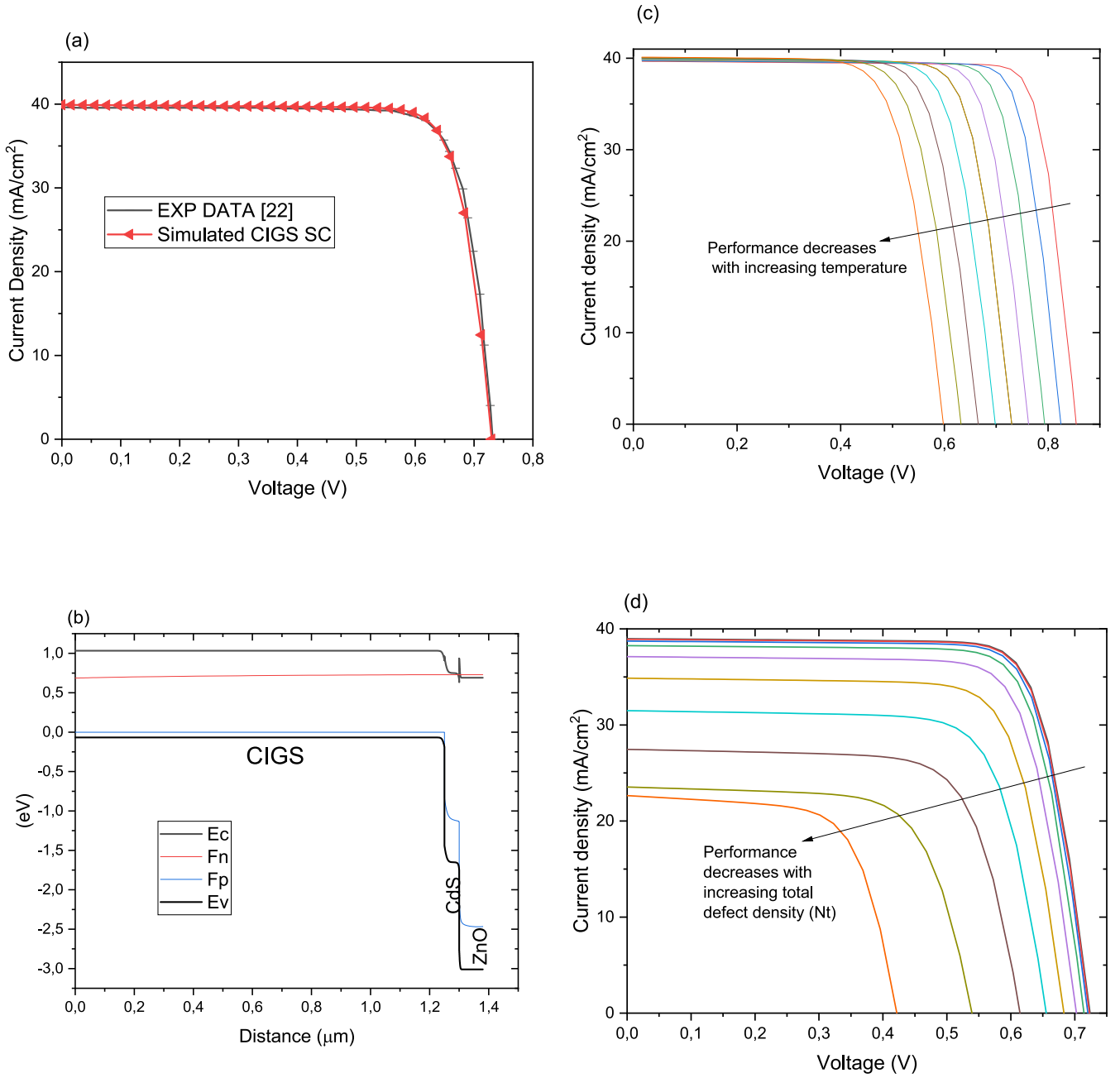


Fig. 2. (a) I-V characteristics of experimental data and simulation results associated with the conventional CdS/CIGS solar cell, (b) Energy band diagram of the CIGS solar cell, (c) J-V characteristics of the CIGS solar cell with different temperatures, (d) J-V characteristics of the CIGS solar cell affected by different total defect densities (N_t).

alignment of both curves in Fig. 2(a).

In the Fig. 2(b), the energy band diagram for a CIGS solar cell illustrates the conduction band (E_c), valence band (E_v), and quasi-Fermi levels (F_n and F_p) across the CIGS, CdS, and ZnO layers. The CIGS absorber's bandgap aligns well with the CdS buffer layer, enabling efficient electron transfer to the ZnO transparent conducting oxide (TCO), which also collects electrons and allows light transmission. A valence band offset between CIGS and CdS confines holes within the CIGS layer, reducing interface recombination. The quasi-Fermi level separation in the CIGS layer indicates photo-generated potential driving the current, while the gradient in the quasi-Fermi levels reflects an electric field that aids charge carrier transport. The ZnO layer's high conduction band energy minimizes electron recombination at the front surface. In CIGS solar cells, the open-circuit voltage decreases with increasing

temperature, attributed to elevated recombination rates induced by higher thermal energy, as reflected by the leftward shift of the curves. Elevated temperatures can also reduce the fill factor, potentially due to increased series resistance or reduced charge carrier collection efficiency under thermal stress. Although the short-circuit current density may exhibit a slight increase with temperature due to bandgap narrowing, this effect is typically minor compared to the pronounced degradation in V_{oc} , as illustrated in Fig. 2(c). Fig. 2(d) presents the J-V curve, demonstrating the impact of increasing total defect density N_t on the performance of the CIGS SC. As N_t increases, the J-V curves shift leftward, signifying a reduction in open-circuit voltage, likely due to elevated recombination at defect sites [25]. Higher defect densities introduce additional pathways for carrier recombination before reaching the contacts, decreasing the effective output voltage. The curves also

exhibit a reduced “square” shape with increasing N_t , indicating a drop in the fill factor. High defect density values may contribute to higher series resistance or hinder charge carrier collection, further impairing the FF. Additionally, at higher N_t , the short-circuit current density decreases marginally, as defects can act as traps that restrict the transport of photogenerated carriers. It is noteworthy that SCAPS-1D simulator inherently incorporates non-radiative recombination processes, including Shockley-Read-Hall (SRH) recombination through defects present in both the bulk material and interfaces. These processes are incorporated in the simulations through parameters such as defect density, capture cross-sections, and defect energy levels, which influence the recombination dynamics in the investigated device. The specific properties of the materials used for the calibration are included in the Table.1.

To learn about the impact of different configurations on the analyzed device, Fig. 3 depicts how the characteristics and properties of the CIGS solar cell layers influence the overall efficiency of the solar cell.

The absorber layer plays a critical role in photovoltaic devices, serving as the primary semiconductor material with a direct band gap and a high absorption coefficient. It captures most of the incident sunlight near the p-n junction. This is demonstrated in Fig. 3(a), where an increase in shallow acceptor density and thickness of the absorber layer leads to a notable improvement in solar cell efficiency. In the other hand, the buffer layer is crucial in forming the p-n junction and facilitating charge carrier separation in chalcogenide CIGS solar cells [33–35]. As shown in Fig. 3(b), the influence of the buffer layer on cell efficiency primarily arises from its shallow donor density, with a less pronounced but negative effect related to its thickness. This can be explained by the fact that a thinner buffer layer permits a greater portion of photons to pass through without being absorbed, thus allowing more photons to reach the absorber layer, where they can generate additional electron-hole pairs, enhancing the overall performance of the solar cell.

Fig. 3(c) presents a contour plot of solar cell efficiency as a function of varying buffer layers and CIGS absorber layer thickness. It is observed that the efficiency improves with increasing absorber layer thickness. This enhancement can be attributed to the reduced likelihood of photon-generated carrier recombination at the back-contact as the thickness grows. Notably, recombination rates are strongly influenced by the junction depth, with deeper junctions minimizing carrier loss and promoting better device performance. The junction depth decreases with respect to the layer thickness as the layer thickness increases. Therefore, at higher absorber layer thickness, the photogenerated carriers are collected more efficiently. The effect of absorber layer thickness occurs regardless of how the type of buffer layer is changed.

In this simulation, it was investigated that the solar cell efficiency decreases with increasing ZnO layer thickness, as shown in Fig. 3(d). The maximum efficiency was observed when the thickness of the ZnO layer was reduced to 0.01 μm . ZnO has a band gap of 3.7 eV, significantly higher than that of the CIGS layer, which is 1.200 eV. As the ZnO layer becomes thinner, it absorbs less photon energy, allowing more low-energy photons to be absorbed by the CIGS absorber layer, thereby

enhancing efficiency. It’s important to highlight that the influence of the ZnO layer is relatively minor and independent of the type of buffer layer used.

The overall performance of CIGS solar cells is predominantly driven by their design parameters, which will be explored using a machine learning-based approach, leveraging a large dataset formed for this purpose.

3. Joint machine learning with deep learning analysis

This section focuses on the integration of machine learning and deep learning techniques for solar cell performance analysis. The first subsection explores feature importance to identify key variables crucial for optimizing solar cell performance. The second subsection applies classification-based deep learning methods to categorize solar cell efficiency into three distinct categories: Good, Worst, and Extremely Worst. Basically, we use both machine learning and deep learning techniques to analyze and optimize the performance of CIGS solar cells in a multi-faceted approach. Machine learning is primarily applied for feature importance analysis through random forest regression with permutation importance, identifying the most influential design parameters for optimizing key figures of merit such as Voc, Jsc, FF, and overall efficiency. This feature importance analysis provides insights into the parameters most critical to improving solar cell efficiency. Additionally, deep learning is employed in the classification stage, where we categorize solar cell efficiencies into distinct groups, Good, Worst, and Extremely Worst, based on efficiency metrics. This classification leverages deep learning’s strength in managing complex patterns within the dataset, further refining our understanding of efficiency outcomes and their associated design parameters. Together, these approaches enable a comprehensive evaluation of parameter impact and performance classification, significantly supporting the research objectives of performance prediction and optimization.”.

3.1. Machine learning-based feature importance analysis

The dataset used in this study comprises 256 instances and focuses on various features related to the performance of thin-film CIGS solar cells. The input features include both categorical and continuous variables, such as Buffer layer, which has six possible material categories (CdS, WS₂, WS₃, ZnMgO, ZnS, and ZnSe), as well as the CIGS shallow acceptor density, ranging from $1\text{E} + 14$ to $1.4\text{E} + 18$ $1/\text{cm}^3$, and the p-CIGS thickness, varying between 500 nm and 1650 nm. Other features include ZnO thickness, Buffer donor density, and Buffer thickness, which significantly affect the electrical and optical properties of the solar cells. Key performance metrics, such as Voc (0.194 to 1.042 V), Jsc (2.91 to 43.06 mA/cm²), FF (15.59 % to 79.42 %), and Efficiency (1.998 % to 23.09 %), provide a detailed evaluation of the cells’ electrical characteristics and overall energy conversion efficiency (see Table.2).

It is noteworthy that the relatively small dataset of 256 instances presents inherent challenges for model generalization, potentially

Table 1
Simulation parameters of different materials reported in the literature.

Parameters	CIGS [26]	ZnO [26,27]	CdS [28]	ZnMgO [26]	ZnS [29]	ZnSe [30,31]	WS ₂ [31,32]
Th (μm)	1.25	0.08	0.05	0.05	0.05	0.05	0.05
Eg(eV)	1.1	3.7	2.4	3.34	3.19	2.9	1.8
χ (eV)	4.25	3.9	4.2	3.7	4.1	4.09	3.95
ϵ	13.6	9	10	9	10	10	13.6
N_C (cm ⁻³)	2.2×10^{18}	2.2×10^{18}	2.2×10^{18}	2.2×10^{18}	2.9×10^{18}	2.2×10^{18}	1.0×10^{18}
N_V (cm ⁻³)	1.8×10^{19}	1.8×10^{19}	1.8×10^{19}	1.8×10^{19}	2.0×10^{19}	1.8×10^{19}	2.4×10^{19}
μ_n (cm ² /V.s)	100	80	100	100	45	400	100
μ_p (cm ² /V.s)	25	330	25	25	10	110	100
N_D (cm ⁻³)	0	1×10^{18}	1×10^{18}	8×10^{18}	1.55×10^{17}	1×10^{18}	1×10^{18}
N_A (cm ⁻³)	1.4×10^{18}	0	0	0	0	0	0

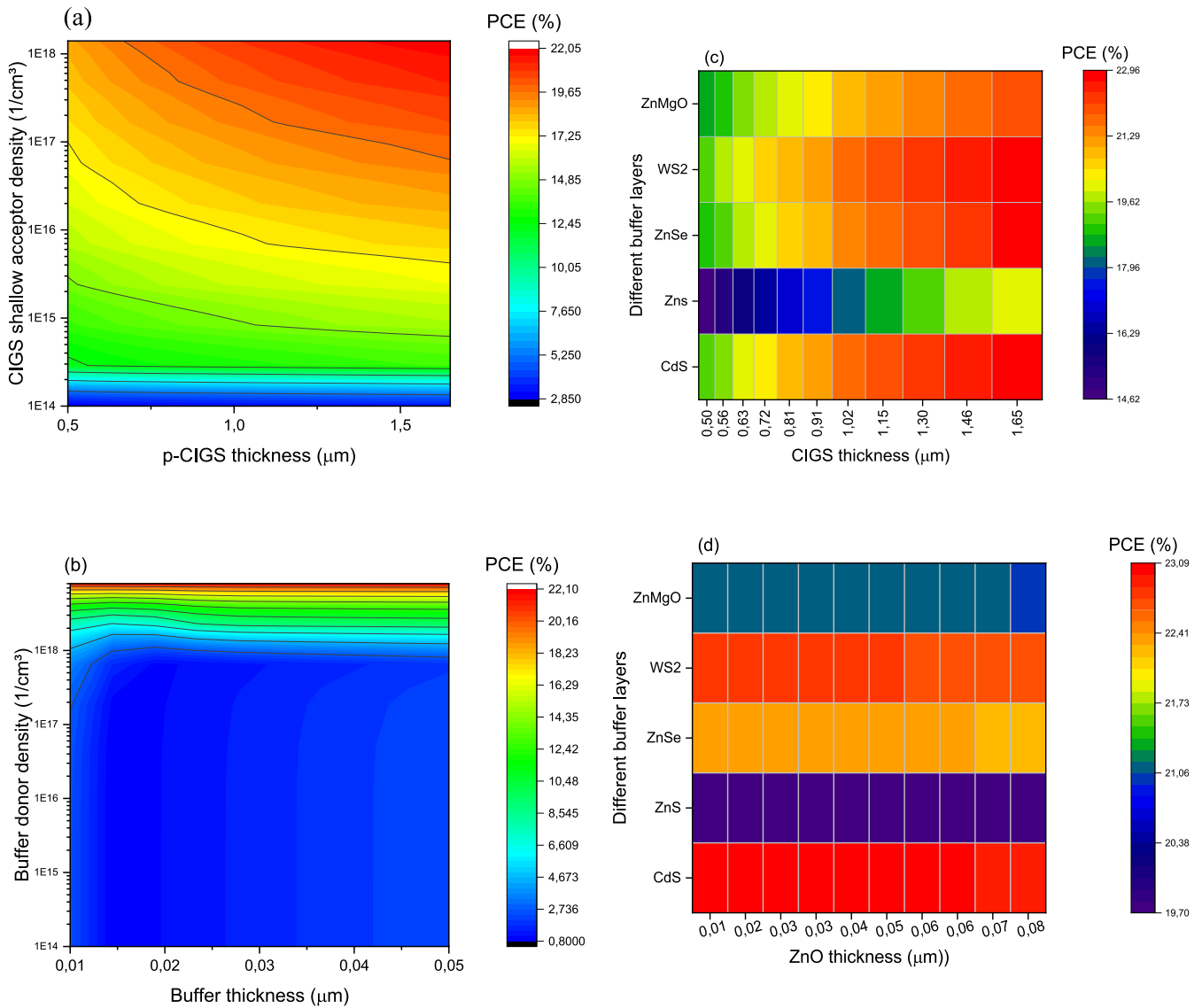


Fig. 3. Efficiency contour plot showing influence of (a) CIGS layer thickness and shallow acceptor density, (b) buffer layer thickness and shallow donor density, (c) buffer layer variation with CIGS layer thickness, (d) buffer layer variation with ZnO layer thickness.

Table 2

Overview of dataset used.

Dataset features	Categories	Min	Max
Buffer	CdS, WS2, WS3, ZnMgO, ZnS, ZnSe	—	—
CIGS shallow acceptor density (1/cm³)	—	1×10^{14}	$1,4 \times 10^{18}$
p-CIGS thickness (cm)	—	5×10^{-5}	$1,65 \times 10^{-4}$
ZnO thickness (cm)	—	1×10^{-6}	8×10^{-6}
Buffer donor density (1/cm³)	—	1×10^{14}	8×10^{18}
Buffer thickness (cm)	—	1×10^{-6}	5×10^{-6}
Recorded FoMs: Voc (V)	—	0,1941	1,0420
Jsc (mA/cm²)	—	2,9128	43,0570
FF (%)	—	15,5892	79,4219
Efficiency (%)	—	1,9976	23,0938

leading to overfitting where the model captures specific patterns rather than underlying distributions. To mitigate this, we employed Bayesian optimization targeting multiple hyperparameters, with a particular emphasis on L2 regularization. This systematic approach not only

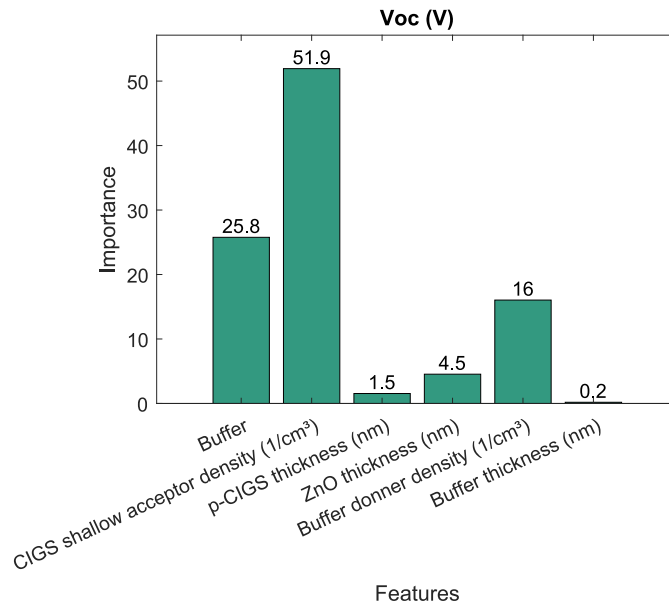
enhances model robustness but also improves generalization to unseen data by effectively reducing overfitting. Additionally, we utilized SMOTE (Synthetic Minority Oversampling Technique) to address class imbalance, further strengthening the training process. While incorporating a larger dataset would indeed provide additional validation and potentially enhance the system's performance, we believe our current methodology demonstrates a strong commitment to rigor and reliability, ensuring that the findings are both robust and applicable within the given constraints. Future work will explore extending the dataset and leveraging advanced validation techniques such as extensive cross-validation to reinforce these results.

In feature importance analysis, understanding how various features affect specific outputs such as Voc, Jsc, FF, and overall efficiency is essential for optimizing solar cell performance. In this study, machine learning models are not assumed to inherently affect feature importance. Rather, it evaluates each feature's impact by measuring changes in the objective function approximation error when feature values are shuffled. The feature importance scores, therefore, reflect the model's sensitivity to each feature rather than any modification or influence on the features themselves. In this analysis, a random forest regression model with permutation importance was used to evaluate the impact of

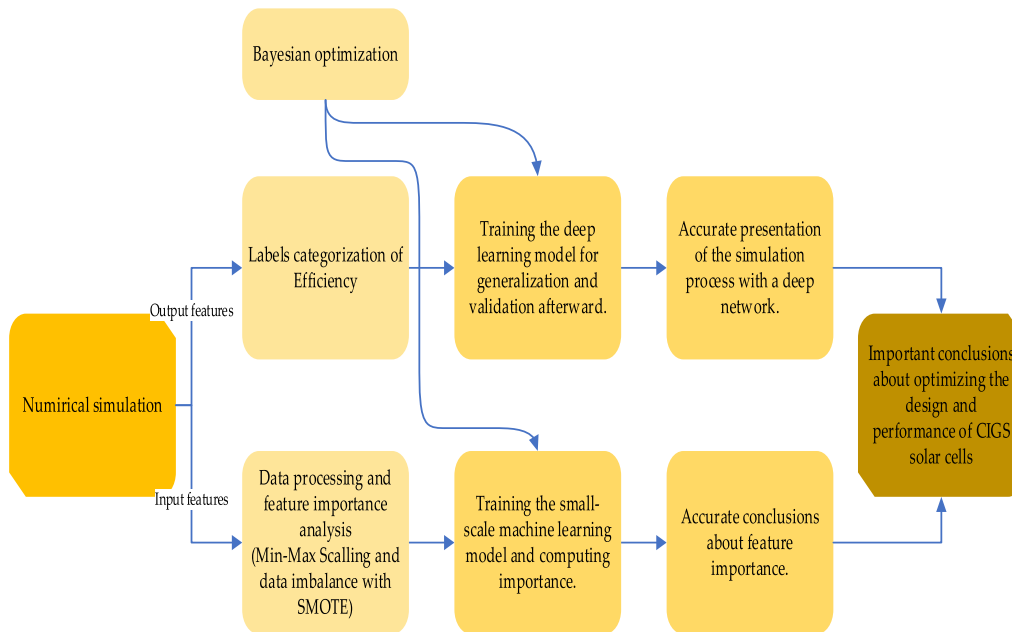
each feature on the output variables, as described in previous studies [36–38]. To ensure the model performs at its best, Bayesian optimization was employed to fine-tune the hyperparameters of the random forest, maximizing its predictive accuracy and efficiency. Figs. 4-7 bar charts illustrating the impact of various features on solar cell performance metrics, including Voc (V), Jsc (mA/cm²), FF (%), and Efficiency (%). Each chart represents the feature importance derived from random forest regression analysis, highlighting the relative influence of each feature on the respective performance metrics.

Starting with Voc, where results are showcased in Fig. 4(a), the buffer shows a substantial positive impact, indicating that the buffer material is critical in maintaining a higher potential difference across

the cell. The CIGS shallow acceptor density has the strongest positive effect on Voc, which suggests that an increased acceptor density enhances the built-in electric field, thus significantly boosting Voc. Other factors, such as the p-CIGS thickness and ZnO thickness, show smaller positive effects on Voc, while buffer donor density and thickness contribute moderately, indicating that the physical and material properties of the buffer layer are essential for improving the voltage. Moreover, Fig. 4(b) presents a comprehensive diagram illustrating the overall contributions of the study. It highlights the importance of various features and how deep learning techniques play a pivotal role in evaluating and enhancing the performance of the device efficiency. The diagram provides a detailed view of how each feature influences key parameters



(a)



(b)

Fig. 4. Importance of both the study and feature importance on device efficiency: (a) Impact of various features on Voc (V); (b) Impact of various study contributions on device efficiency.

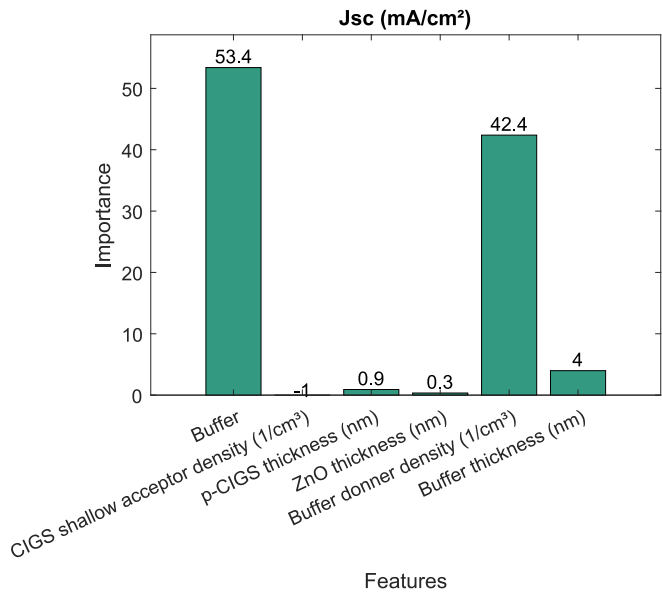


Fig. 5. Impact of various features on Jsc(mA/cm²).

and how machine learning-driven analysis contributes to optimizing the device’s performance.

For Jsc, and as addressed in Fig. 5, the buffer again plays a crucial role, with the highest positive influence on current density, indicating its importance in charge collection and recombination reduction. Interestingly, the CIGS shallow acceptor density has a negative effect on Jsc, implying that while it boosts Voc, it may hinder current density due to potential recombination issues or reduced carrier mobility. The p-CIGS and ZnO thicknesses have minimal positive effects, whereas buffer donor density shows a strong positive impact, indicating that higher

donor density facilitates better charge transport, leading to improved current density. Buffer thickness, while positive, plays a less significant role compared to donor density.

As illustrated in Fig. 6, and in terms of FF, the buffer continues to show a major positive influence, highlighting its contribution to optimizing charge collection efficiency. The CIGS shallow acceptor density has a moderate positive impact on FF, while the p-CIGS and ZnO thicknesses contribute only marginally. Buffer donor density, however, plays a strong role in improving the FF, suggesting that optimizing donor

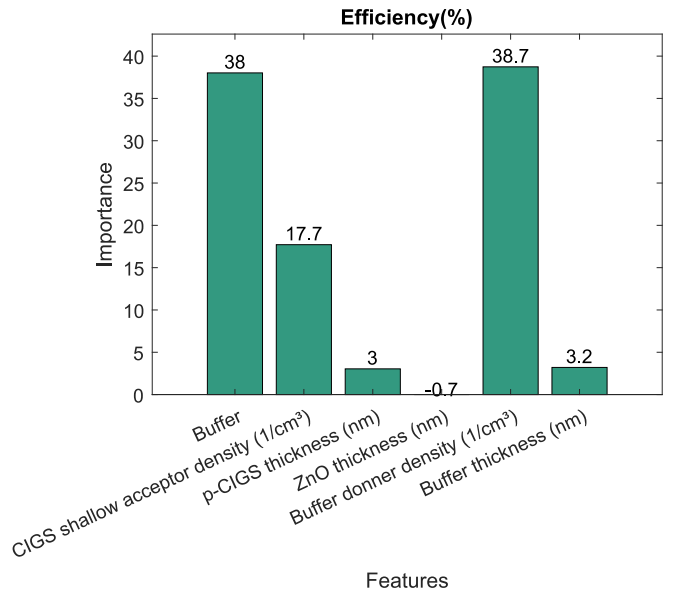


Fig. 7. Impact of various features on Efficiency (%).

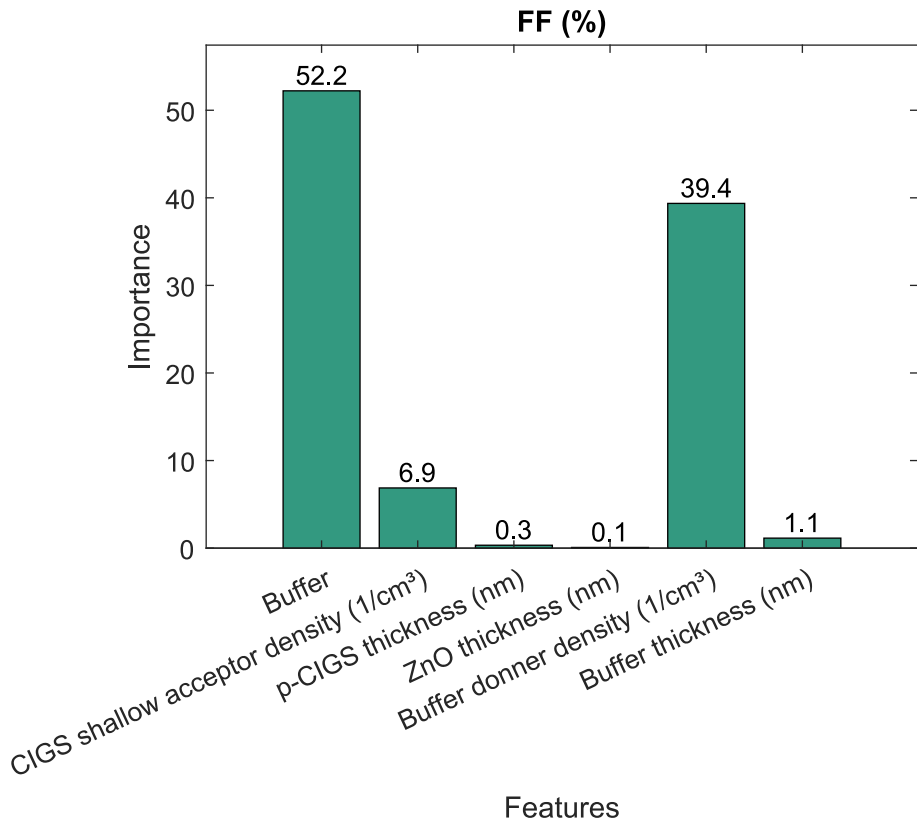


Fig. 6. Impact of various features on FF (%).

concentration can significantly enhance the overall power conversion capabilities. Buffer thickness again shows a minor positive impact on FF, reinforcing that the buffer's material and structural properties are key to performance.

When it comes to efficiency, and following the results depicted in Fig. 7, the buffer emerges as the most critical factor, with a strong positive effect across all performance indicators (Voc, Jsc, and FF), making it the single most influential variable in improving solar cell efficiency. The CIGS shallow acceptor density, while beneficial for Voc, has mixed effects on Jsc and FF, limiting its overall contribution to efficiency. Notably, ZnO thickness exhibits a slightly negative effect on efficiency, possibly due to losses introduced at higher thicknesses, despite moderate positive contributions to Voc and Jsc [38,39]. Buffer donor density proves to be nearly as influential as the buffer itself, showing that optimizing donor density is essential for maximizing efficiency. Finally, buffer thickness has a smaller but positive effect on efficiency, suggesting that while it contributes to overall performance, its impact is less pronounced compared to other buffer-related properties.

In conclusion, buffer-related variables, particularly buffer thickness and buffer donor density, emerge as the most impactful contributors to overall solar cell efficiency, demonstrating a consistently high positive influence across key performance metrics, Voc, Jsc, FF, and PCE. This comprehensive feature importance analysis underscores the essential role of buffer properties in supporting charge transport, enhancing voltage potential, and optimizing charge collection, which collectively elevate solar cell performance. By focusing on these buffer-related factors, we address the most critical variables affecting efficiency gains. Meanwhile, variables such as ZnO and p-CIGS thickness exhibit more nuanced, sometimes mixed effects, contributing positively in specific cases but potentially limiting efficiency if not carefully balanced. As suggested, our analysis combines and discusses these features collectively to provide a holistic view of their relative importance, demonstrating that while buffer features offer the most substantial efficiency gains, careful optimization of ZnO and p-CIGS thickness remains essential to maximize performance without risking diminishing returns.

3.2. Deep learning for efficiency classification

To categorize the efficiency values, we utilized specific provided statistics and applied specific mathematical formulas to establish thresholds. We categorize efficiency values based on standard statistical thresholds derived from the mean and standard deviation of the dataset, following a methodology similar to statistical binning or z-score-based classification often used in performance assessment and risk categorization (e.g., [40]). This approach allows us to distinguish efficiency values relative to statistical benchmarks, with "Good" representing values more than one standard deviation above the mean, "Worse" within one standard deviation, and "Extremely Worse" more than one standard deviation below the mean. The statistics included a minimum efficiency $E_{min} = 2\%$, a maximum efficiency $E_{max} = 23.09\%$, a mean value $\mu_E = 16.98\%$, and a standard deviation value $\delta_E = 5.90\%$. Based on these, we defined the subsequent categorization strategy. Efficiency values significantly lower than the mean are categorized as "Extremely Worse." We determined the threshold T for this category by subtracting one standard deviation from the mean, and then rounding them to nearest multiple m as in (4). Therefore, any efficiency value less than or equal to 11.08 is classified as "Extremely Worse."

$$T_{\text{Extremelyworse}} = \left\lfloor \frac{(\mu_E - \delta_E) + \frac{m}{2}}{m} \right\rfloor \quad (4)$$

Efficiency values falling within one standard deviation below and above the mean are categorized as "Worse." The lower bound of this category is the same as the threshold for "Extremely Worse," which is 11.08. The upper bound is calculated by adding one standard deviation to the mean

as in (5). Therefore, efficiency values between 10 and 20 (inclusive of 10 and exclusive of 20) are categorized as "Worse."

$$T_{\text{Worse}} = \left\lceil \frac{(\mu_E + \delta_E) + \frac{m}{2}}{m} \right\rceil \quad (5)$$

Efficiency values significantly higher than the mean are classified as "Good." This threshold is set as any value greater than one standard deviation above the mean. Therefore, any efficiency value exceeding 20 is categorized as "Good."

After defining the categorization thresholds, we proceeded with data normalization using min-max normalization to ensure that all input features are scaled to a uniform range before being fed into the subsequent steps. This step is crucial for achieving consistent performance across the model by standardizing the input scale.

Fig. 8(a) and (b) provide a comprehensive overview of the dataset prepared for analysis. Fig. 8(a) illustrates the class proportions within the dataset, offering insights into the distribution of different categories. Fig. 8(b) visualizes the dataset in 2 dimensions using t-SNE (t-distributed Stochastic Neighbor Embedding), showcasing how the classes are distributed and separated in a reduced feature space. This visualization helps in understanding the clustering of data points and the separability of different classes within the dataset. Fig. 8 reveals two main data complexities that need to be addressed. First, the dataset exhibits significant class imbalance, which necessitates balancing strategies to ensure fair representation. Second, the t-SNE scatter plot demonstrates the dataset's complexity, showing that data points from different classes are heavily intertwined. This overlap indicates the need for advanced techniques to manage the intricate relationships and distinguish between classes effectively. Accordingly, the class imbalance issue is addressed using the SMOTE [41], which produces a more balanced class distribution. Fig. 8(c) and (d) illustrates the results of this balancing process, showcasing both class proportions and scatter plots of the balanced data. The class proportions appear well-balanced, indicating effective resampling. However, the scatter plots reveal that samples from different classes still exhibit some degree of overlap, suggesting areas where further refinement or more advanced techniques may be needed to improve class separation.

The second challenge, which involves complex class overlaps, is addressed using a multilayer deep neural network specifically designed to handle intricate relationships within the data [42]. To effectively manage this complexity, the network incorporates a Long Short-Term Memory (LSTM) layer, as shown in Fig. 9. LSTMs are particularly well-suited for this task because they are capable of capturing long-term dependencies and patterns in sequential data, which helps in distinguishing between overlapping classes. The network architecture also includes an input layer, dropout layer, and output layer. Hyperparameters are optimized through Bayesian optimization, covering aspects such as the number of hidden units, maximum epochs, mini-batch size, initial learning rate, gradient threshold, L2 regularization, dropout rate, recurrent dropout rate, state activation function, gate activation function, and sequence length.

LSTM unit, includes an input gate i_t , a forget gate f_t , a cell State C_t , and an output gate o_t [43]. These gates and cells manage the influx of new input, decide which information from the previous cell state to discard, retain long-term data, and control the extent of cell state output, and are calculated according to (6–11). x_t denotes the input at time instant t , h_{t-1} represents the hidden state of $t-1$, δ stands for a sigmoid activation function, $\tanh$ refers to a hyperbolic tangent activation function, W and U are weight matrices, b is the bias vector, and $\odot$ denotes element-wise multiplication.

$$i_t = \sigma(W_i x_t + U_i h_{t-1} + b_i) \quad (6)$$

$$f_t = \sigma(W_f x_t + U_f h_{t-1} + b_f) \quad (7)$$

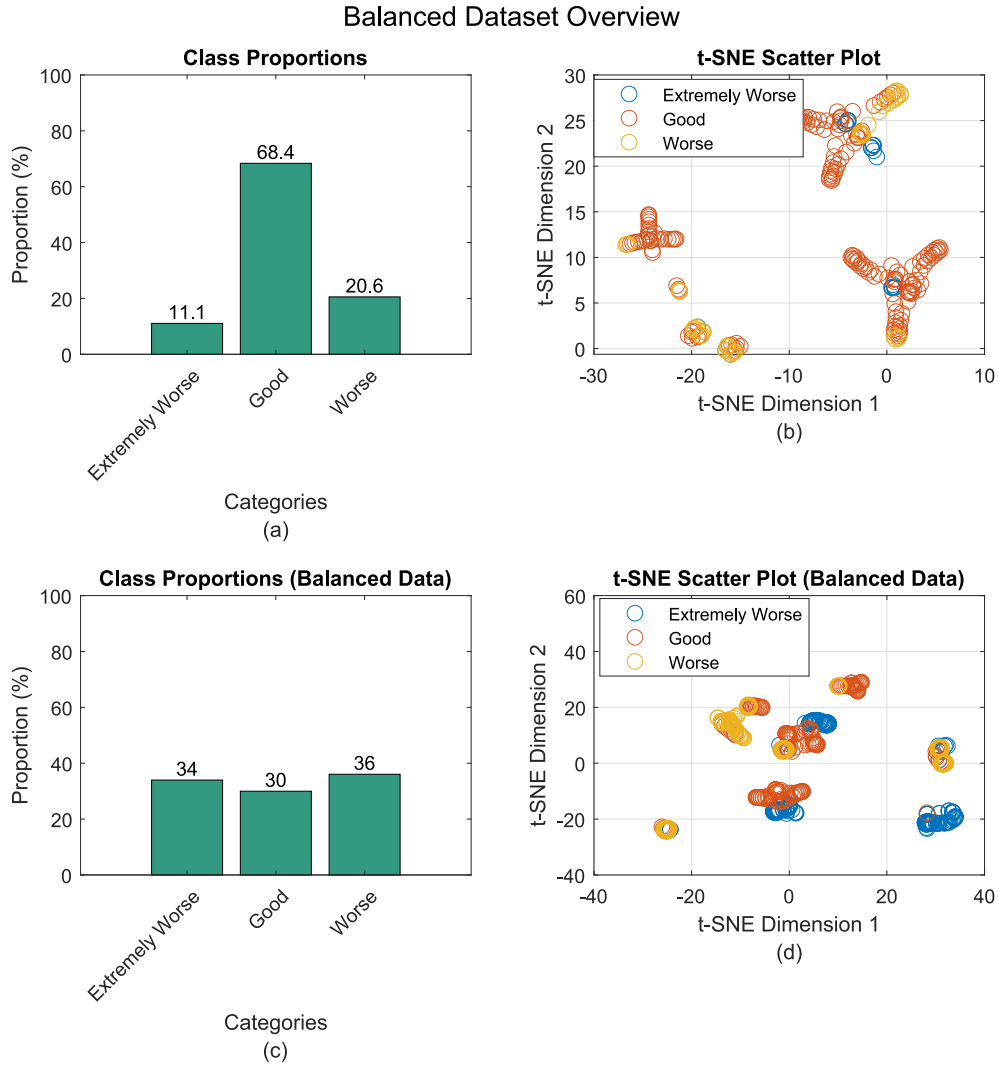


Fig. 8. Visualization of the dataset before and after balancing: (a) Class proportions of raw dataset; (b) t-SNE scatter plot of raw dataset; (c) Class proportions of balanced dataset; (d) t-SNE scatter plot of balanced dataset.

$$o_t = \sigma(W_o x_t + U_o h_{t-1} + b_o) \quad (8)$$

$$\tilde{C}_t = \tanh(W_c x_t + U_c h_{t-1} + b_c) \quad (9)$$

$$C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t \quad (10)$$

$$h_t = o_t \odot \tanh(C_t) \quad (11)$$

The deep learning model is trained using a 60 %-40 % data split. We recognize that the training dataset may appear relatively small, with such a training-to-test split. However, this allocation was chosen intentionally to establish an initial, proof-of-concept model using the available data. Our goal was to explore and demonstrate the model's potential in identifying key patterns and trends in a foundational way. While a larger dataset would ideally enhance training robustness, we believe this setup provides a meaningful starting point, showing feasibility and generating insights that can guide future research as more data becomes accessible.

The evaluation process begins with an analysis of the model's learning behavior, focusing on the loss function to understand how well the model is learning during training. Following this, the evaluation proceeds with a detailed examination of ROC curves and confusion matrices to assess the model's performance. Finally, a comprehensive set

of classification evaluation metrics is analyzed to provide a thorough assessment of the model's accuracy and effectiveness.

The loss function in Fig. 10 exhibits a notable pattern throughout the training epochs. Initially, from epochs 0 to 40, there is a slight convergence, indicating early stabilization of the model as it starts to learn. Following this, from epochs 45 to 50, the loss shows minor fluctuations, suggesting that the model is adjusting to nuances in the data and might be experiencing some variability in its learning process. From epochs 50 to 400, the loss function demonstrates a smooth and continuous convergence. This trend reflects a more stable and effective learning phase, where the model gradually refines its performance with only a few intermittent spikes. These fluctuations, while present, are relatively minor and do not significantly disrupt the overall convergence trend. This behavior implies that the model achieves a stable learning phase after an initial period of adjustment. The initial convergence and subsequent fluctuations indicate that the model is navigating through its training space, adjusting to the data's complexity. The smooth convergence in the latter stages highlights effective learning and generalization, with occasional spikes being indicative of minor adjustments or overfitting risks. Overall, this pattern suggests that the model is performing well and steadily improving its accuracy over time.

The ROC curve in Fig. 11 showcases a right angle and an AUC of 1, which means that the model perfectly distinguishes between the positive and negative classes, achieving a 100 % true positive rate and a 0 % false

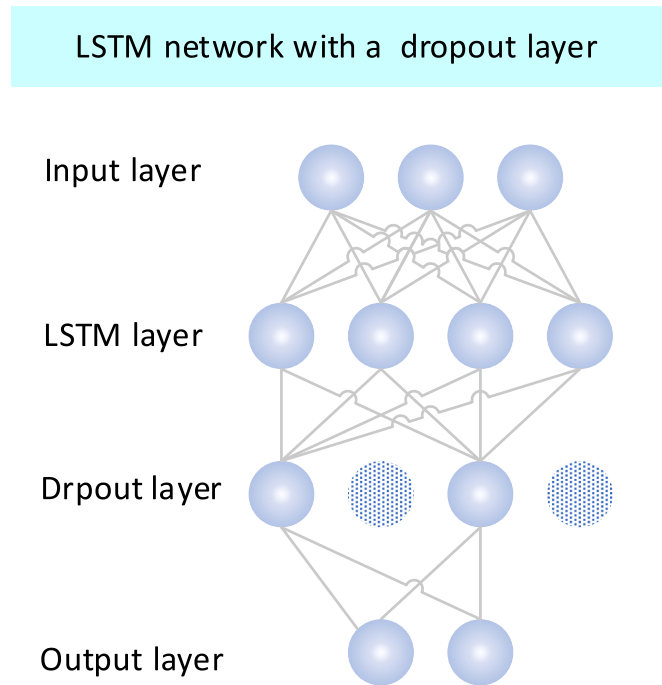


Fig. 9. Overview of utilized deep learning network.

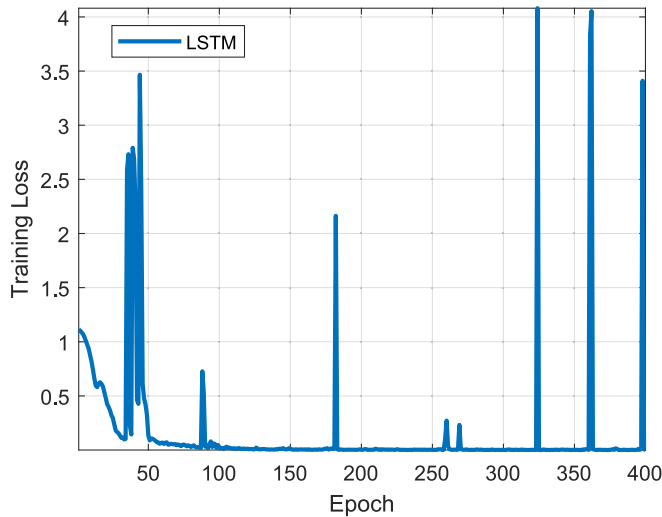


Fig. 10. Loss function behavior of deep learning model.

positive rate. This indicates optimal performance with no overlap between the classes, reflecting perfect classification accuracy.

The confusion matrix in Fig. 12 illustrates the effectiveness of the classification model by highlighting the proportion of correctly and incorrectly classified instances. It shows a slight misclassification where instances labeled as “Extremely worse” are incorrectly predicted as “Good” with an accuracy of 5.3 %. This indicates that the model has some difficulty distinguishing between these two classes, suggesting that further refinement or additional features may be needed to enhance the model’s performance in correctly classifying “Extremely worse” cases.

The final evaluation of the model’s performance, as detailed in Table.3, shows that it achieved perfect scores in accuracy, precision, recall, and F1 score during training, indicating flawless classification on the training data with no misclassifications. The training process was completed in approximately 8.38 s, reflecting an efficient training time. For the testing phase, the model demonstrated high performance with an

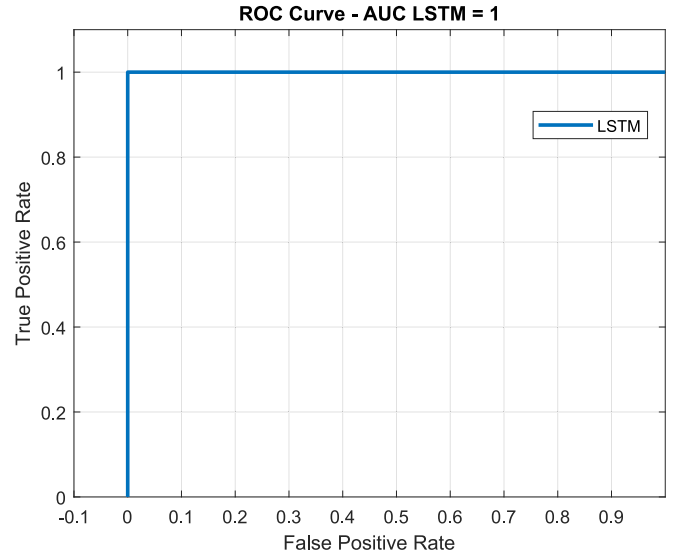


Fig. 11. ROC curves of deep learning model.

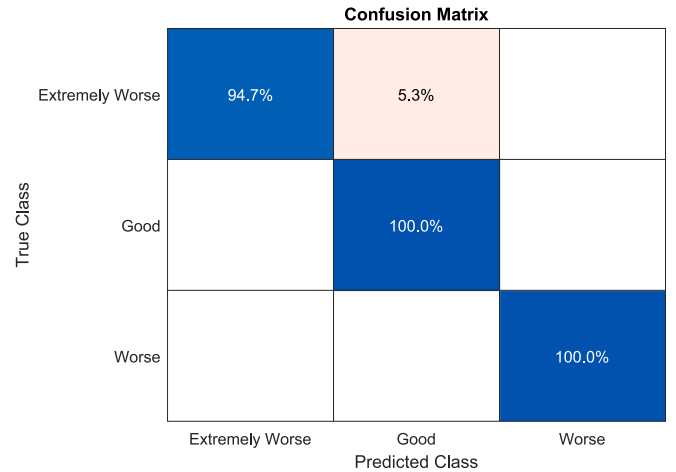


Fig. 12. Confusion matrix results.

Table 3
Obtained numerical results.

Training	Accuracy	1
	Precision	1
	Recall	1
	F1 score	1
Testing	Accuracy	0.9805
	Precision	0.9808
	Recall	0.9825
	F1 score	0.9811
Training	Time (s)	8.3828

accuracy of 98.05 %, precision of 98.08 %, recall of 98.25 %, and an F1 score of 98.11 %. These results highlight the model’s strong generalization capabilities and its effectiveness in classifying unseen data. The slight decrease in performance from training to testing is typical and suggests minor overfitting, which is a common occurrence. This can be addressed through further model optimization or regularization techniques. Overall, the model’s performance metrics underscore its robustness and reliability in practical applications, confirming its ability to deliver accurate and consistent results.

Bayesian optimization was employed to tune hyperparameters in our deep learning model, specifically targeting L2 regularization, learning rates, and other parameters of network architecture and activation parameters. This method uses a probabilistic model, typically a Gaussian Process, to make informed decisions about which hyperparameter configurations to evaluate next, balancing exploration of new configurations with exploitation of those that have previously shown promise. As we iteratively evaluated different sets of hyperparameters, the surrogate model was updated based on previous results, allowing us to refine our search effectively. This systematic approach led to the identification of hyperparameter configurations that significantly enhanced model accuracy and reduced overfitting, resulting in a more robust model that generalizes better to unseen data and ultimately improves our predictive capabilities. Table.4 outlines the optimal learning hyperparameters determined through Bayesian optimization for training our deep learning model. The model features 106 hidden units to effectively capture intricate data patterns. We have selected a maximum of 847 epochs to ensure adequate training time for achieving convergence. A mini batch size of 96 strikes a balance between computational efficiency and precise gradient estimation. The initial learning rate is set to 0.071208, facilitating moderate weight adjustments during optimization. To ensure training stability, a gradient threshold of 0.30609 is employed. L2 regularization is configured at 0.014012 to mitigate overfitting by penalizing excessively large weights. Dropout rates are set at 0.15559 for general dropout and 0.20338 for recurrent dropout, which helps prevent overfitting by randomly omitting neurons during training. The state activation function is tanh, introducing non-linearity to enhance the model's ability to learn complex patterns, while the gate activation function is hard-sigmoid, managing the flow of information within recurrent layers. Lastly, a sequence length of 21 is utilized to process data in 21-time step chunks, effectively capturing temporal dependencies.

In the context of our research on CIGS solar cells, it is essential to situate our contributions within the existing body of work, which is relatively scarce. Our study specifically addresses the challenges associated with predicting the efficiency of CIGS solar cells using advanced methodologies. Accordingly, if we compare our work to existing methods and recent machine learning or optimization studies on CIGS solar cells, certain conclusions emerge [17,44,45]. These insights highlight both advancements and limitations, providing a clearer perspective on how our approach contributes to the field. For instance, in comparison to the study in [42], which uses random forest and gradient boosting on experimental datasets to predict key factors for CIGS solar cell efficiency, our approach integrates simulation-based data with both machine learning (RF regression for feature importance) and deep learning for efficiency classification. This dual approach, combined with Bayesian optimization, enhances predictive accuracy and categorizes efficiency levels more granularly. While the study in [42] is based on experimentally driven data, offering high practical relevance, our simulation data expands possible scenario analyses, though it may require further validation for experimental applicability.

4. Conclusion

This paper presents a novel design methodology that integrates SCAPS-1D numerical simulation with machine learning (ML) to investigate critical design parameters for CIGS thin-film solar cells (TFSCs). Through comprehensive numerical simulations, we successfully reproduce the I-V characteristics of the CIGS solar cell, analyzing the influence of key design parameters on output FoMs such as open-circuit voltage (Voc), short-circuit current (Jsc), fill factor (FF), and power conversion efficiency (PCE). The ML approach includes data collection, preprocessing, feature extraction, and training of supervised learning models. After evaluating various models, the optimal one is selected for its predictive accuracy. The ML-driven methodology efficiently predicts and identifies ideal design parameters, demonstrating its powerful

Table 4
Learning hyperparameters.

Parameter Name	Value
Number of Hidden Units	106
Maximum Epochs	847
Mini Batch Size	96
Initial Learning Rate	0.0712
Gradient Threshold	0.3060
L2 Regularization	0.0140
Dropout Rate	0.1555
Recurrent Dropout Rate	0.2033
State Activation Function	tanh
Gate Activation Function	hard-sigmoid
Sequence Length	21

potential in prediction and classification. Our findings show that ML integration reduces both cost and development time while potentially enhancing device performance. By incorporating additional theoretical and experimental data, machine learning can generate more accurate and reliable results. The proposed design framework shows an efficient and reliable strategy for optimizing both the design and fabrication processes of thin-film solar cells (TFSCs).

CRediT authorship contribution statement

A. Maoucha: Writing – original draft, Software, Investigation, Formal analysis. **T. Berghout:** Writing – original draft, Software, Investigation, Formal a, nalysis, Data curation. **F. Djeflal:** Writing – review & editing, Validation, Supervision, Methodology, Investigation, Conceptualization. **H. Ferhati:** Software, Investigation, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

[1] P.K. Nayak, S. Mahesh, H.J. Snaith, D. Cahen, Photovoltaic solar cell technologies: analysing the state of the art, *Nature Review Materials* 4 (2019) 269–285.
[2] V. Tyag, et al., Progress in Solar PV technology: Research and achievement, *Renewable and Sustainable Reviews* 20 (2013) 443–461.
[3] M.A. Green, et al., Solar cell efficiency tables (Version 60), *Progress in Photovoltaics* 30 (2022) 687–701.
[4] K. Yoshikawa, et al., Silicon heterojunction solar cell with interdigitated back contacts for a photoconversion efficiency over 26%, *Nat. Energy* 2 (2017) (2017) 17032.
[5] H. Ferhati, F. AbdelMalek, F. Djeflal, “ Improved PCE in Stable Lead-Free Perovskite Solar Cells Based on Band Engineering of ETL and Absorber Solar Energy” vol. 262 (2023) 111805.
[6] T. Ablekim, et al., Thin-film solar cells with 19% efficiency by thermal evaporation of CdSe and CdTe, *ACS Energy Lett.* 5 (2020) 892–896.
[7] H. Ferhati, F. Djeflal, Exceeding 30% efficiency for an environment-friendly tandem solar cell based on earth-abundant Se/CZTS materials, *Physica e. Low-Dimensional Systems and Nanostructures* 109 (2019) 52–58.
[8] K. Kacha, et al., Efficiency improvement of CIGS solar cells using RF sputtered TCO/Ag/TCO thin-film as prospective buffer layer, *Ceramics International* 48 (2022) 20194–20200.
[9] H. Ferhati, F. Djeflal, F. AbdelMalek, Towards improved efficiency of SnS solar cells using back grooves and strained-SnO2 buffer layer: FDTD and DFT calculations, *Journal of Physics and Chemistry of Solids* 178 (2023) 111353.
[10] J.Y. Cho, et al., Achieving over 4% efficiency for SnS/CdS thin-film solar cells by improving heterojunction interface quality, *J. Mater. Chem. A* 8 (2020) 20658–20665.
[11] B. Augustine, et al., Recycling perovskite solar cells through inexpensive quality recovery and reuse of patterned indium tin oxide and substrates from expired devices by single solvent treatment, *Solar Energy Materials and Solar Cells* 194 (2019) 74–82.
[12] SILVACO TCAD (2012).
[13] I.I. Malek, H. Imtiaz, S. Subrina, Machine learning driven performance enhancement of perovskite solar cells with CNT as both hole transport layer and back contact, *Solar Energy* 278 (Aug. 2024) 112737.

- [14] N. Shrivastav, et al., Exploring KGeCl₃ material for perovskite solar cell absorber layer through different machine learning models, *Solar Energy* 278 (2024) 112784.
- [15] I.I. Malek, et al., Machine learning driven performance enhancement of perovskite solar cells with CNT as both hole transport layer and back contact, *Solar Energy* 278 (2024) 112737.
- [16] H. Ferhati, T. Berghout, A. Benyahia, F. Djeflal, Machine Learning DFT-Based Approach to Predict the Electrical Properties of Tin Oxide Materials. The 10th International Electronic Conference on Sensors and Applications, 2023.
- [17] V.C. Karade, et al., Machine Learning Assisted Analysis, Prediction, and Fabrication of High-Efficiency CZTS_{Se} Thin Film Solar Cells, *Advanced Functional Materials* 33 (41) (Jun. 2023).
- [18] H. Ferhati, F. Djeflal, Graded band-gap engineering for increased efficiency in CZTS solar cells, *Optical Materials* 76 (2018) 393–399.
- [19] H. Ferhati, F. Djeflal, Performance assessment of TCO/metal/TCO multilayer transparent electrodes: from design concept to optimization, *Journal of Computational Electronics* 19 (2020) 815–824.
- [20] A. Maoucha, F. Djeflal, H. Ferhati, F. AbdelMalek, Eco-friendly perovskite/CZTS_{Se} tandem cell exceeding 28% efficiency through current matching and bandgap optimization: a numerical investigation, *The European Physical Journal plus* 138 (7) (Jul. 2023).
- [21] A. Maoucha, F. Djeflal, and H. Ferhati, “Numerical Investigation of a New Double-Absorber Lead-free Perovskite Solar Cell via SCAPS-1D,” Oct. 2023.
- [22] A. Maoucha, F. Djeflal, H. Ferhati, SCAPS-FDTD simulation of 20.1 % efficient Perovskite-SnS tandem solar cell based on alternative charge transport layers and Au-nanoparticles, *Physica Scripta*, Dec. (2023).
- [23] A. Maoucha, F. Djeflal, T. Berghout, H. Ferhati, Photoresponsivity Enhancement of SnS-Based Devices Using Machine Learning and SCAPS Simulations, *Engineering Proceedings* 58 (2023) 128.
- [24] J. Hwang, et al., Improved carrier transport in CIGS solar cells induced by Ag treatment, *Journal of Alloys and Compounds* 886 (2021) 161193.
- [25] S. Taheri, A.A. kordbacheh, M. Minbashi, A. Hajjiah, Effect of defects on high efficient perovskite solar cells, *Optical Materials* 111 (2020) 110601.
- [26] F.S. Ahmadpanah, A.A. Orouji, I. Gharibshahian, Improving the efficiency of CIGS solar cells using an optimized p-type CZTS_{Se} electron reflector layer, *Journal of Materials Science Materials in Electronics* 32 (2021) 22535–22547.
- [27] I. Gharibshahian, A.A. Orouji, S. Sharbati, “suitable Top Cell Partners for Copper Indium Gallium Selenide-Based Tandem Solar Cells to Achieve >30% Efficiency”, *Physica Status Solidi (a)* vol. 218 (2021) 2000796.
- [28] S. Mishra, K. Bhargava, D. Deb, Numerical simulation of potential induced degradation (PID) in different thin-film solar cells using SCAPS-1D, *Solar Energy* 188 (2019) 353–360.
- [29] T. Chargui, F. Lmai, K. Rahmani, Advancements in CIGS/ZnS Heterojunction Solar Cells: Experimental and Numerical Analysis, *Optik* 314 (2024) 172008.
- [30] K.P. Mishra, B.K. Pandey, S. Pandey, Unveiling the Potential of FASnI₃ Solar Cells through Advanced Charge Transport Materials: A SCAPS-1D Perspective, *Journal of Alloys and Compounds* 1006 (2024) 176283.
- [31] H. Karmaker, A. Siddique, B.K. Das, M.N. Islam, Modeling and performance investigation of novel inorganic Cs₄CuSb₂Cl₁₂ nanocrystal perovskite solar cell using SCAPS-1D, *Results in Engineering* 22 (2024) 102106.
- [32] K.J. Abbas, A. Bahrani, Investigating the potential of WO₃ and WS₂ as Cd-free buffer layers in Sb₂Se₃-Based thin-film solar cells: A numerical study with SCAPS-1D software, *Solar Energy Materials and Solar Cells* 272 (2024) 112891.
- [33] M. Sood, et al., Electrical barriers and their elimination by tuning (Zn, Mg)O buffer composition in Cu(In, Ga)S₂ solar cells: systematic approach to achieve over 14% power conversion efficiency, *J. Phys. Energy* 3 (2022) 045005.
- [34] S.M. Shamim, et al., “Design, performance analysis and efficiency optimization of copper indium gallium selenide (CIGS) solar cell,” *European Scientific Journal*, vol.11(6), 2015.
- [35] A. Maoucha, H. Ferhati, F. Djeflal, F. AbdelMalek, Highly efficient Cd-Free ZnMgO/CIGS solar cells via effective band-gap tuning strategy, *Journal of Computational Electronics* 22 (3) (2023) 887–896.
- [36] S. Zhang, “Permutation Tests for the Partial Coefficient in a Multiple Linear Regression”, *Asia-Pacific Conf. Appl. Math.* (2023).
- [37] L. Lausser, R. Szekeley, H.A. Kestler, Permutation-invariant linear classifiers, *Mach. Learn.* (2024).
- [38] P.S. Mbule, T.H. Kim, B.S. Kim, H.C. Swart, O.M. Ntwaeaborwa, Effects of particle morphology of ZnO buffer layer on the performance of organic solar cell devices, *Solar Energy Materials and Solar Cells* 112 (2013) 6–12.
- [39] A.R. bin Mohd Yusoff, H.P. Kim, J. Jang, High performance organic photovoltaics with zinc oxide and graphene oxide buffer layers, *Nanoscale* 6 (2014) 1537–1544.
- [40] S. Kuttathammakul, J. Smeyers-Verbeke, D.L. Massart, D. Coomans, S. Noack, The mean and standard deviation of data, some of which are below the detection limit: an introduction to maximum likelihood estimation, *TrAC Trends in Analytical Chemistry* 19 (2000) 215–222.
- [41] N.V. Chawla, K.W. Bowyer, L.O. Hall, W.P. Kegelmeyer, “SMOTE: Synthetic minority over-sampling technique”, *Jun, J. Artif. Intell. Res.* 16 (2) (2002) 321–357.
- [42] T. Berghout, M. Benbouzid, M.A. Ferrag, Multiverse Recurrent Expansion With Multiple Repeats: A Representation Learning Algorithm for Electricity Theft Detection in Smart Grids, *IEEE Trans. Smart Grid* 14 (6) (Nov. 2023) 4693–4703.
- [43] F. Kadri, et al., Neural classification method in fault detection and diagnosis for voltage source inverter in variable speed drive with induction motor, in: *IEEE Eighth International Conference and Exhibition on Ecological Vehicles and Renewable Energies (EVER)*, 2013, pp. 1–5, <https://doi.org/10.1109/EVER.2013.6521549>, 2019.
- [44] V.C. Karade, et al., Machine Learning Aided Optimization of P1 Laser Scribing Process on Indium Tin Oxide Substrates, *Advanced Intelligent Systems* 6 (2024) 2300733.
- [45] V.C. Karade, et al., Unraveling the Effect of Compositional Ratios on the Kesterite Thin-Film Solar Cells Using Machine Learning Techniques, *Crystals* 13 (Nov. 2023) 1581.